

OPENER: Open Partitioning Embedding for Named Entity Recognition

Anonymous Author(s)

Paper submitted for double-blind review

LyRICS Symposium 2026

Abstract

Open-world named entity recognition (NER) must assign entity types that were not fixed at training time. Recent systems either train a dedicated encoder or prompt billion-parameter language models, and they are compared almost exclusively on accuracy, ignoring their compute and energy cost. We present **OPENER**, a lightweight open-world NER pipeline assembled from off-the-shelf components: a frozen mention detector, a Matryoshka-truncatable sentence embedder sharpened by a light contrastive fine-tuning, and a balanced linear classifier for entity typing, so that no encoder is trained from scratch. We benchmark OPENER against GLiNER, GNER and OWNER on 13 datasets along three axes at once: quality, measured by Adjusted Mutual Information (AMI), inference latency, and energy consumption. Our analysis separates entity typing on gold mentions, where the contrastive embedding space is highly discriminative, from the end-to-end setting, where the dominant open-world bottleneck is shown to be mention detection rather than typing. The energy-aware comparison further exposes order-of-magnitude efficiency gaps between approaches that a quality-only evaluation would hide. On gold mentions, OPENER’s balanced linear typing, its embedding space sharpened by error-driven hard-negative mining, surpasses a zero-shot transfer of OWNER (62.5 vs 43.0 AMI); end-to-end, where mention detection becomes the shared bottleneck, OPENER still attains the best AMI of the compared open-world systems (40.2), at 1.6 Wh per run, an order of magnitude or more below the contrastive-clustering (14.3 Wh) and generative (36.5 Wh) baselines. We release the code, the per-experiment configurations and the full benchmark.

Index Terms

open-world named entity recognition, entity typing, contrastive representation learning, Matryoshka embeddings, frugal artificial intelligence, energy-efficient machine learning, adjusted mutual information

I. INTRODUCTION

[Placeholder - introduction to be written.]

II. RELATED WORK

OPENER sits at the intersection of three lines of research: open-world and zero-shot named entity recognition (NER), pretrained entity representations refined by contrastive learning, and energy-aware machine learning. We review each in turn and state how OPENER differs.

A. Closed-set and Span-based NER

Traditional NER is cast as supervised sequence labeling over a fixed inventory of entity types, typically with a pretrained encoder followed by a token- or span-level classifier. Span-based formulations enumerate candidate spans and classify them directly, which simplifies the handling of nested and discontinuous entities [1]. These models reach high accuracy on in-domain benchmarks such as CoNLL-2003 [2], but their label set is frozen at training time. Adding a new entity type requires re-annotating data and retraining the model, which is the central limitation that open-world NER seeks to remove.

B. Open-world and Zero-shot NER

Open-world NER aims to recognise entity types that were not fixed at training time. GLiNER [3] treats the candidate type names as queries to a bidirectional encoder, a DeBERTaV3 backbone [4], and scores every span against them in a single forward pass, offering a fast alternative to autoregressive tagging. GNER [5] instead casts NER as text generation with a sequence-to-sequence model [6] and shows that explicitly modeling negative, non-entity instances sharpens type boundaries. Closest to our work, Genest et al. [7] propose OWNER, an unsupervised open-world pipeline that decouples mention detection from entity typing: it embeds detected mentions with an encoder refined by contrastive learning and then clusters them into dynamically named types. Other approaches prompt a general-purpose large language model to perform extraction directly, as in ChatIE [8], or focus on recovering entities that detectors tend to miss [9]. OPENER adopts the decoupled detect-then-type view of OWNER, but replaces the trained encoder and the unsupervised clustering with a pretrained, Matryoshka-truncatable embedding space and a balanced linear classifier, so that no encoder is trained from scratch.

C. LLM-based and Few-shot NER

A complementary line distills or instruction-tunes large language models for open NER. UniversalNER [10] distills a chat model into a smaller LLaMA-based student [11] over tens of thousands of entity types, while GoLLIE [12] conditions a code language model on annotation guidelines to follow unseen schemas. Few-shot methods reduce supervision further through prompt optimization [13], knowledge-graph-guided contrastive learning [14], or token-level contrastive objectives such as CONTaiNER [15]. These methods are accurate but rest on models with billions of parameters, whose inference cost, latency and energy use are substantial, and which may not even fit on commodity hardware. This cost is precisely what motivates the lightweight design and the energy-aware evaluation of OPENER.

D. Entity Representations and Contrastive Learning

OPENER relies on a pretrained sentence embedder rather than a task-specific encoder. Nomic Embed [16] provides an open, long-context model trained with the Matryoshka objective [17], whose representations can be truncated to lower dimensions without retraining, trading quality for compute. Such embedders build on the Sentence-BERT line of work [18], and we sharpen the entity-type geometry with a triplet margin objective in the spirit of FaceNet [19]. Embedding labels and entities in a shared space [20], modeling entity types with Gaussian components [21], generating synthetic entity-bearing text for augmentation [22], and using mixtures to augment training data [23] are recurring ideas in low-resource NER, all grounded in classical Gaussian mixture models fitted by expectation-maximization [24], [25]. Unlike methods that train a dedicated encoder, OPENER keeps the pretrained space largely frozen and only applies a light contrastive fine-tuning, which tests whether a general-purpose embedding is already discriminative enough for entity typing.

E. Efficient and Green AI

A growing body of work argues that efficiency should be reported alongside accuracy. Strubell et al. [26] quantify the financial and environmental cost of training NLP models, Schwartz et al. [27] advocate efficiency as a first-class evaluation criterion, and Patterson et al. [28] analyse the carbon footprint of machine-learning training and how to reduce it. Tools such as CodeCarbon [29] make per-run energy and carbon estimates practical. Open-world NER systems, however, are still compared almost exclusively on quality. OPENER instead reports a three-axis comparison of quality, measured by Adjusted Mutual Information [30], inference speed, and energy, placing the frugality of each method on an equal footing with its accuracy.

In summary, existing open-world NER methods either train a dedicated encoder [7] or query costly large language models [10], [12], and they are compared on accuracy alone. OPENER fills both gaps: it reuses pretrained components with only a light contrastive step and a balanced linear classifier, and it is evaluated jointly on quality, speed and energy.

III. METHOD

[Placeholder - method to be written once the configuration is frozen.]

IV. EXPERIMENTS

A. Datasets

We evaluate on **13 evaluation sets** that span news, social media, spoken queries, encyclopedic text, biomedical abstracts and manufacturing literature, following the dataset selection of Genest et al. [7]. CrossNER [31] contributes five domain-specific sets, evaluated separately (artificial intelligence, literature, music, politics, science). The remaining eight are CoNLL-2003 [2], WNUT-17 [32], the MIT-Restaurant and MIT-Movie spoken-query corpora [33], FabNER [34], the BioNLP/JNLPBA-2004 biomedical task [35], and the GUM [36] and GENTLE [37] multi-genre corpora. Label granularity ranges from four coarse types in CoNLL-2003 to twelve fine-grained categories in FabNER, including opaque domain acronyms. Two datasets used by [7], GENIA [38] and i2b2, are license-gated and therefore not re-measured here; BioNLP/JNLPBA-2004 serves as our biomedical proxy. To homogenise evaluation cost across systems, we cap each test split at 1000 sentences (the full split is used when it is smaller). Table I summarises the benchmark.

B. Baselines

We compare OPENER against representative open-world NER systems run end-to-end on the same splits. GLiNER [3] is evaluated in its small, medium and large variants to trace a size/efficiency trade-off. GNER [5] is a generative tagger built on T5 [6]; we run the T5-base checkpoint. We further compare against OWNER [7], executed in its original environment. Instruction-tuned and few-shot large-language-model NER systems, namely UniversalNER [10], GoLLIE [12] and ChatIE [8], rely on 7B-parameter backbones [11] that do not fit our hardware budget even under 4-bit NF4 quantization [39]; for these we report the figures published by their authors rather than re-measuring them.

TABLE I
THE 13 EVALUATION SETS. “TEST” IS THE NUMBER OF TEST SENTENCES USED (CAPPED AT 1000).

Dataset	Abbr.	Domain	Test
CrossNER-AI	AI	Encyclopedic (AI)	430
CrossNER-Literature	Lit	Encyclopedic (literature)	416
CrossNER-Music	Mus	Encyclopedic (music)	465
CrossNER-Politics	Pol	Encyclopedic (politics)	651
CrossNER-Science	Sci	Encyclopedic (science)	543
WNUT-17	WNUT	Social media	689
MIT-Restaurant	Rest	Restaurant search	1000
MIT-Movie	Movie	Movie trivia	1000
FabNER	Fab	Manufacturing	1000
BioNLP/JNLPBA-2004	Bio	Biomedical	1000
CoNLL-2003	CoNLL	News	1000
GUM	GUM	Multi-genre	1000
GENTLE	GEN	Multi-genre (challenge)	1000

C. Metrics

Our primary metric is the Adjusted Mutual Information (AMI) [30] between predicted and gold type assignments, following [7], complemented by macro-averaged F_1 and accuracy. Because open-world predictions need not share the gold label vocabulary, AMI is computed over the aligned set of mentions, which makes it well suited to comparing dynamically named types. Beyond quality, we report two efficiency axes: inference *speed*, as the median (p50) latency per sentence, and *energy*, as kilowatt-hours and grams of CO₂eq per run.

D. Evaluation Protocol

Predicted and gold mentions are aligned by exact character offsets. Following [7], an unmatched gold mention (a false negative) and an unmatched predicted mention (a false positive) are each assigned a distinct sentinel label before computing AMI, so that detection errors are penalised rather than ignored. We distinguish two protocols that must not be mixed in the same comparison. In the *typing-on-gold* protocol, the entity-typing component is applied to gold mention spans, which measures typing quality in isolation under perfect detection. In the *end-to-end* protocol, mentions are produced by the detector and the sentinel labels above are active, which is the setting comparable to the baselines. All runs use a single fixed random seed (42); we note this as a limitation, since we do not report variance over seeds.

E. Implementation Details

All experiments run on a single consumer-grade GPU with 6 GB of VRAM (CUDA 12.1, PyTorch 2.5.1 [40]). We use the HuggingFace Transformers library [41] for the baselines, sentence-transformers [18] for the Matryoshka embedder [16], [17], and scikit-learn [42] for the linear classifier, whose solver is LIBLINEAR [43]. Latency is measured with two warmup passes followed by timed passes with GPU synchronisation, reporting p50, p95 and p99 per sentence. Energy is measured with CodeCarbon [29] in offline mode using the French grid carbon intensity; when hardware power counters are unavailable, we fall back to a thermal-design-power estimate and flag the measurement method in the report. All systems are measured on the same machine. Because the laptop-class GPU thermally throttles under sustained load, per-sentence latency and per-run energy carry roughly $\pm 15\%$ run-to-run variance; we therefore read the per-system averages as indicative and do not over-interpret individual per-dataset cells. A controlled-temperature re-measurement of all systems in a single session is left to future work.

F. Main Results

The tables below report the benchmark on all 13 sets. OWNER is run in its native protocol [7]: a single model is trained once on the CoNLL-2003 source (a one-time cost of 88 minutes on our 6 GB GPU) and then applied *zero-shot* to every target set (cells marked §). CoNLL-2003 is therefore in-domain for OWNER (marked †) and is excluded when reading OWNER as a transfer system. One protocol asymmetry must be kept in mind when reading the typing scores: OPENER fits a lightweight linear classifier on each target’s training split (a supervised probe), whereas OWNER, GLiNER and GNER use no target-side labels. The two systems thus trade annotation for inference cost: OPENER needs target labels but stays cheap at test time, while OWNER needs none but pays a heavy per-sentence cost (Table III). To close this asymmetry we also evaluate OPENER-ZS, a zero-shot variant that drops the supervised probe and types each mention by cosine similarity to a label-name prototype, using no target labels (Table VI). End-to-end it matches the zero-shot OWNER baseline (34.2 vs 34.3 mean AMI) while keeping OPENER’s frugal inference profile (137 ms, 1.85 Wh vs 616 ms, 14.3 Wh); on gold mentions it edges OWNER out (44.4 vs 43.0, Table VII). Two label-free refinements push it further: a transductive k-means on the prototypes (36.8 end-to-end), and *fusing in the detector’s own zero-shot label predictions*—a signal the centroid otherwise discards—which reaches 39.4

TABLE II

OPEN-WORLD NER, END-TO-END PROTOCOL. ADJUSTED MUTUAL INFORMATION ($\text{AMI} \times 100$, $\uparrow$). ABBREVIATIONS FOLLOW TABLE I; BOLD MARKS THE BEST SYSTEM PER COLUMN. OPENER USES THE ERROR-DRIVEN HARD-NEGATIVE MINING EMBEDDER SELECTED IN TABLE VI AND OBTAINS THE BEST MEAN AMI OF THE COMPARED SYSTEMS. $\S$ MARKS ZERO-SHOT SYSTEMS THAT USE NO TARGET-SIDE LABELS: OWNER (TRAINED ONCE ON CONLL-2003, THEN APPLIED TO EACH TARGET) AND OPENER-ZS, THE ZERO-SHOT VARIANT OF OPENER THAT TYPES MENTIONS BY NEAREST LABEL-NAME PROTOTYPE INSTEAD OF A TRAINED PROBE; THE PLAIN VARIANT MATCHES OWNER END-TO-END (34.2 VS 34.3) AT AN ORDER OF MAGNITUDE LOWER COST (TABLE III, TABLE IV), AND TWO LABEL-FREE REFINEMENTS—TRANSDUCTIVE PROTOTYPES (36.8) AND FUSION WITH THE DETECTOR’S OWN LABEL PREDICTIONS (39.4)—MAKE OPENER-ZS THE BEST ZERO-SHOT SYSTEM, AHEAD OF GLiNER-L (38.9). $\dagger$ CONLL-2003 IS IN-DOMAIN FOR OWNER (ITS TRAINING SOURCE) AND FOR OPENER (ITS CONTRASTIVE SOURCE), AND IS NOT BOLD AS A TRANSFER RESULT; THE BEST ZERO-SHOT SYSTEM ON CONLL IS GNER. THE TYPING-ON-GOLD COMPARISON IS REPORTED IN TABLE VII.

Model	AI	Lit	Mus	Pol	Sci	WNUT	Rest	Movie	Fab	Bio	CoNLL	GUM	GEN	Avg
GLiNER-S	41.7	49.8	56.4	50.6	53.3	29.7	26.1	35.6	10.7	31.8	18.9	34.1	36.1	36.5
GLiNER-M	42.0	50.1	57.4	50.2	54.2	28.1	27.9	36.4	11.3	34.9	24.9	32.8	31.6	37.1
GLiNER-L	42.9	48.9	58.0	49.8	52.8	28.9	30.9	37.6	28.1	34.6	29.7	32.7	30.5	38.9
GNER	41.9	46.9	53.5	48.9	49.9	30.0	30.0	36.8	17.0	29.1	43.7	32.8	31.6	37.9
OWNER $\S$	43.9	44.0	46.6	50.9	50.3	22.3	7.6	24.0	18.7	26.5	50.3 $\dagger$	29.1	31.3	34.3
OPENER	40.2	49.6	55.3	50.3	51.7	24.9	33.6	37.8	30.2	36.5	44.8 $\dagger$	33.7	34.2	40.2
OPENER-ZS $\S$	31.6	43.8	41.7	36.7	44.2	23.0	28.4	32.4	28.3	29.1	42.6 $\dagger$	31.5	31.2	34.2
OPENER-ZS (transd.) $\S$	33.7	48.9	53.7	45.3	45.9	24.2	30.4	34.7	27.9	27.9	44.4 $\dagger$	31.0	30.8	36.8
OPENER-ZS (fusion) $\S$	39.4	48.8	58.0	48.6	52.8	29.6	31.5	35.5	27.8	35.7	43.1 $\dagger$	31.2	30.4	39.4

TABLE III

INFERENCE LATENCY. MEDIAN PER-SENTENCE LATENCY IN MILLISECONDS ($\downarrow$), END-TO-END. ABBREVIATIONS FOLLOW TABLE I; BEST PER COLUMN IN BOLD. $\S$ OWNER DOES NOT EXPOSE A PER-SENTENCE CALL: ITS FIGURE IS THE TOTAL INFERENCE WALL-CLOCK DIVIDED BY THE NUMBER OF SENTENCES (AN AMORTISED MEAN, NOT A TRUE P50), AND EXCLUDES THE ONE-TIME 88-MIN TRAINING ON CONLL-2003. CONLL ($\dagger$) IS OWNER’S TRAINING SOURCE; ITS LATENCY COMES FROM A SEPARATE INFERENCE-ONLY LOAD-AND-TEST RUN ON THE CONLL TEST SET (THE TRAINED MODEL IS RELOADED, THEN RUN LIKE THE TRANSFER TARGETS), SO IT IS COMPARABLE TO THE OTHER COLUMNS; IT IS IN-DOMAIN (MARKED $\dagger$) AND, AS IN THE AMI TABLES, INCLUDED IN THE 13-SET AVERAGE. OWNER IS AN ORDER OF MAGNITUDE SLOWER THAN GLiNER BECAUSE IT RE-ENCODS EVERY MENTION AND CLUSTERS AT INFERENCE TIME. OPENER-ZS IS THE ZERO-SHOT OPENER VARIANT (LABEL-NAME NEAREST-CENTROID TYPING); ITS HEAD IS A SINGLE COSINE COMPARISON, SO IT REPORTS A GENUINE PER-SENTENCE P50 THAT TRACKS OPENER’S AND STAYS FAR BELOW OWNER’S.

Model	AI	Lit	Mus	Pol	Sci	WNUT	Rest	Movie	Fab	Bio	CoNLL	GUM	GEN	Avg
GLiNER-S	21	22	21	21	21	20	19	20	20	22	20	21	21	21
GLiNER-M	36	36	37	38	39	39	38	40	41	42	44	46	45	40
GLiNER-L	144	149	153	149	155	137	132	145	147	143	131	141	137	143
GNER	3431	5408	7461	6472	7110	2906	1610	3344	4310	5034	2914	3435	1500	4226
OWNER $\S$	500	712	895	920	757	311	281	532	438	558	719 $\dagger$	759	626	616
OPENER	222	271	293	159	132	100	92	98	104	108	89	100	94	143
OPENER-ZS	109	135	156	172	175	136	127	137	133	136	117	127	124	137
OPENER-ZS (transd.)	104	130	149	158	186	186	217	233	194	132	75	95	98	151
OPENER-ZS (fusion)	218	254	291	277	289	185	173	205	84	98	92	104	97	182

end-to-end, the best of all compared zero-shot systems, ahead of GLiNER-L (38.9) and within 0.8 AMI of the supervised probe, all without any target label. The supervised probe remains the stronger configuration (40.2 end-to-end, 62.5 on gold), so OPENER-ZS is best read as the frugal, annotation-free operating point of the same pipeline.

G. Ablations

We justify each architectural choice with a one-factor-at-a-time search. Starting from the default pipeline, we vary a single component while holding the others fixed, and keep the setting with the highest mean AMI. Among embedder variants, an error-driven *hard-negative mining* pass, fine-tuning the contrastive embedder on the entity pairs it most often confuses (e.g. *writer/person*, *album/band*, *country/location*), mixed with correctly-typed mentions for diversity, raises typing-on-gold AMI from 54.0 to 60.1; scaling the mined triplets (3000 $\rightarrow$ 8000) and epochs (2 $\rightarrow$ 3) lifts it further to 62.5 (+8.5 over the contrastive embedder, better on 12 of 13 sets), our selected configuration. Table VI reports this search; the protocol comparison between typing-on-gold and end-to-end is discussed alongside the main results. The remaining axes confirm the rest of the pipeline: frozen Nomic embeddings are weakly discriminative for typing (25.8 AMI), so the contrastive step is what makes the space usable (+28.2); the detector trades quality for cost (GLiNER-S $\rightarrow$ L: 37.7 $\rightarrow$ 40.2 AMI at 51 $\rightarrow$ 143 ms end-to-end); and for typing a balanced linear SVM edges out a per-class GMM and logistic regression (62.5 vs 56.1 vs 55.6). Replacing the supervised typing head with a label-name nearest-centroid removes target supervision entirely, at a cost of 18.1 AMI (62.5 $\rightarrow$ 44.4); this zero-shot variant (OPENER-ZS) still surpasses the zero-shot OWNER baseline (43.0, Table VII) while fitting in 293ms instead of 3.5s, making it the frugal choice when no target labels are available. A label-free *transductive* refinement of the prototypes—spherical k-means seeded by the label names and run once on the unlabelled test mentions—recovers 7.0 AMI on gold mentions (44.4 $\rightarrow$ 51.4, Table VI) and 2.6 end-to-end (34.2 $\rightarrow$ 36.8, Table II, above OWNER’s 34.3),

TABLE IV

ENERGY. ENERGY PER RUN IN WATT-HOURS ($\downarrow$), END-TO-END. ABBREVIATIONS FOLLOW TABLE I; BEST PER COLUMN IN BOLD. § OWNER IS MEASURED AT INFERENCE ONLY (ZERO-SHOT TRANSFER); ITS ONE-TIME CONLL-2003 TRAINING (88 MIN) IS AMORTISED ACROSS ALL TARGETS AND NOT CHARGED PER RUN. CONLL ($\dagger$) IS OWNER’S TRAINING SOURCE; ITS ENERGY COMES FROM A SEPARATE INFERENCE-ONLY LOAD-AND-TEST RUN ON THE CONLL TEST SET (22.0 WH), COMPARABLE TO THE OTHER COLUMNS; IT IS IN-DOMAIN (MARKED $\dagger$) AND INCLUDED IN THE 13-SET AVERAGE.

OWNER REMAINS ENERGY-HEAVY BECAUSE EACH SENTENCE TRIGGERS PER-MENTION ENCODING AND CLUSTERING. OPENER-ZS IS THE ZERO-SHOT OPENER VARIANT; ITS PER-RUN ENERGY TRACKS OPENER’S (THE COSINE TYPING HEAD IS NEGLIGIBLE) AND STAYS AN ORDER OF MAGNITUDE BELOW OWNER’S.

Model	AI	Lit	Mus	Pol	Sci	WNUT	Rest	Movie	Fab	Bio	CoNLL	GUM	GEN	Avg
GLiNER-S	0.19	0.17	0.20	0.27	0.22	0.27	0.35	0.39	0.40	0.42	0.37	0.41	0.40	0.31
GLiNER-M	0.31	0.29	0.33	0.48	0.40	0.50	0.66	0.76	0.77	0.79	0.69	0.82	0.75	0.58
GLiNER-L	0.68	0.67	0.77	1.04	0.91	1.01	1.37	1.51	1.54	1.49	1.38	1.52	1.43	1.18
GNER	17.13	24.28	39.75	50.88	44.76	20.60	17.02	37.17	49.11	63.50	45.22	38.69	26.50	36.51
OWNER§	6.6	9.0	12.7	18.3	12.6	6.5	8.6	16.3	13.4	17.1	22.0 $\dagger$	23.2	19.1	14.3
OPENER	1.10	1.26	0.87	1.76	1.41	1.31	1.75	1.87	1.96	1.97	1.73	1.91	1.78	1.59
OPENER-ZS	0.95	1.06	1.37	1.97	1.69	1.52	2.07	2.28	2.15	2.30	2.10	2.33	2.26	1.85
OPENER-ZS (transd.)	0.97	1.04	1.36	1.89	2.04	2.15	3.15	2.68	2.08	1.97	1.48	1.85	1.90	1.89
OPENER-ZS (fusion)	1.14	1.23	1.58	2.08	1.82	1.41	1.82	2.23	1.62	1.87	1.63	1.90	1.79	1.70

TABLE V

SUMMARY: QUALITY VERSUS EFFICIENCY, AVERAGED OVER THE 13 DATASETS. AMI ($\times 100$, $\uparrow$); MEDIAN PER-SENTENCE LATENCY (MS), ENERGY (WH) AND CARBON (GCO₂EQ) PER RUN ($\downarrow$); AND PARAMETER COUNT. § OWNER IS ZERO-SHOT (TRAINED ONCE ON CONLL-2003 FOR 88 MIN, THEN APPLIED TO EVERY TARGET); ITS LATENCY IS AN AMORTISED WALL-CLOCK MEAN (TABLE III); AMI, LATENCY AND ENERGY ARE ALL 13-SET MEANS, OWNER’S IN-DOMAIN CONLL INFERENCE COST BEING MEASURED BY A SEPARATE LOAD-AND-TEST RUN (MARKED $\dagger$ IN TABLE III, TABLE IV) WHILE ITS ONE-TIME 88-MIN TRAINING IS AMORTISED AND NOT CHARGED PER RUN. END-TO-END OPENER PAIRS THE GLiNER-L DETECTOR (330M) WITH THE 137M EMBEDDER, HENCE 467M; ON GOLD MENTIONS THE DETECTOR IS NOT USED (137M). END-TO-END OWNER LIKewise SUMS A DeBERTA-v3-BASE DETECTOR (≈ 184 M) AND A BERT-BASE TYPING ENCODER (110M), HENCE ≈ 294 M; GLiNER-S/M/L SIZES ARE FROM THE PROJECT NOTES AND UNVERIFIED. OPENER-ZS IS THE ZERO-SHOT OPENER VARIANT (NO TARGET LABELS; LABEL-NAME NEAREST-CENTROID TYPING); IT KEEPS OPENER’S 467M FOOTPRINT AND PER-SENTENCE COST YET MATCHES OWNER’S ZERO-SHOT QUALITY (34.2 VS 34.3 AMI) AT 137 MS AND 1.85 WH, I.E. AT A FRACTION OF OWNER’S 616 MS / 14.3 WH.

Model	Params	AMI	p50 (ms)	Energy (Wh)	CO ₂ (g)
GLiNER-S	50M	36.5	21	0.31	0.02
GLiNER-M	200M	37.1	40	0.58	0.03
GLiNER-L	330M	38.9	143	1.18	0.07
GNER	220M	37.9	4226	36.51	2.05
OWNER§	294M	34.3	616	14.3	0.74
OPENER	467M	40.2	143	1.59	0.09
OPENER-ZS	467M	34.2	137	1.85	0.10
OPENER-ZS (transd.)	467M	36.8	151	1.89	0.11
OPENER-ZS (fusion)	467M	39.4	182	1.70	0.10

widening OPENER-ZS’s margin over OWNER while still using no target labels; this is a transductive rather than inductive setting, as it exploits the unlabelled test distribution. We further probed whether the zero-shot head could be improved by aligning entity mentions and label *names* in a shared space, trained on the large label vocabulary of Pile-NER [10] (≈ 4.6 k types) with an entity $\leftrightarrow$ label-name contrastive objective (Table VIII). This produces a *strictly* domain-agnostic embedder (unlike the selected hard-mining embedder, it never sees any target-side data), yet it does not improve mean AMI (42.2 vs 44.4): the Pile-NER vocabulary lifts macro- F_1 (+3.8) and helps encyclopedic domains (CrossNER-Politics +14.2, -Science +8.6) but hurts specialised schemas (MIT-Restaurant −15.9, FabNER −11.8, BioNLP −11.8). A second label-name hard-mining pass degrades it further (39.7 AMI): Pile-NER’s GPT-generated types include many near-synonyms (e.g. *person/per*, *location/loc*), so mining “confusions” between them manufactures negatives that pull apart labels that should stay close. We therefore retain the target-tuned hard-mining embedder for OPENER-ZS and report the Pile-NER variants as a negative ablation. A separate efficiency axis emerges in the *training* cost (Table VI, Fit): per-sentence inference is governed by the detector alone (truncating the Matryoshka dimension or swapping the classifier leaves it unchanged), whereas fitting the typing head is $7\times$ cheaper at dimension 64 than at 768 (for only a 4.6-point AMI drop, $62.5 \rightarrow 57.9$) and $4.5\times$ cheaper for balanced logistic regression than for the linear SVM, a speed/quality knob orthogonal to inference. To isolate the entity-typing component itself, Table VII compares OPENER against OWNER on gold mentions, where detection errors are removed.

TABLE VI

OPENER ARCHITECTURE SEARCH. ONE COMPONENT IS VARIED AT A TIME AROUND THE DEFAULT PIPELINE (**BOLD**). AMI AND MACRO- F_1 ($\times 100$, $\uparrow$); PER-RUN LATENCY, ENERGY AND CARBON ($\downarrow$). * ENTITY-EMBEDDER ROWS ARE TYPING-ON-GOLD OVER ALL 13 SETS; *hard-negative mining* FINE-TUNES THE CONTRASTIVE EMBEDDER A *second* TIME ON THE MENTION PAIRS IT MOST OFTEN CONFUSES (HARD NEGATIVES IN THE SENSE OF [19]), MIXED WITH CORRECTLY-TYPED MENTIONS FOR DIVERSITY (65% HARD NEGATIVES; THE TWO ROWS USE 3000 TRIPLETS / 2 EPOCHS AND, SELECTED, 8000 TRIPLETS / 3 EPOCHS); ERRORS ARE MINED ON EACH TRAINING SPLIT BY CROSS-VALIDATION, SO NO TEST LABEL IS SEEN. $\dagger$ ENTITY-TYPING SWEEP ON ALL 13 SETS (TYPING-ON-GOLD) WITH THE SELECTED HARD-MINING EMBEDDER, SO ITS LINEARSVC (BALANCED) ROW EQUALS THE SELECTED ENTITY-EMBEDDER ROW; $\S$ DIMENSION SWEEP ON ALL 13 SETS (TYPING-ON-GOLD) WITH THE SELECTED HARD-MINING EMBEDDER, TRUNCATING THE MATRYOSHKA REPRESENTATION AFTER ENCODING, SO ITS 768 ROW EQUALS THE SELECTED EMBEDDER AND TYPING ROWS. EFFICIENCY (P50 LATENCY, ENERGY, CO₂) FOLLOWS THE PROTOCOL OF EACH AXIS. ON THE MENTION-DETECTOR AXIS IT IS THE DEPLOYED END-TO-END PER-SENTENCE COST AND VARIES WITH THE DETECTOR (51 $\rightarrow$ 143 MS). ON THE EMBEDDER, DIMENSION AND TYPING AXES (TYPING-ON-GOLD, DETECTION BYPASSED) IT IS THE EMBEDDER-PLUS-TYPING INFERENCE COST MEASURED ON THE 13 TEST SETS: IT IS CONSTANT AT 42 MS / 0.63 WH / 0.04 G BECAUSE THESE CONFIGURATIONS SHARE THE SAME 137M EMBEDDER FORWARD PASS—TRUNCATING THE MATRYOSHKA DIMENSION SLICES THE STORED VECTOR AFTER ENCODING, AND SWAPPING THE LINEAR CLASSIFIER IS NEGLIGIBLE. PER-ROW MEASUREMENT CONFIRMED THIS CONSTANCY, THE RESIDUAL RUN-TO-RUN SPREAD (34 $\rightarrow$ 71 MS ON THE EMBEDDER AXIS) BEING THERMAL THROTTLING ($\pm 15\%$). FIT (MS) IS THE TYPING-HEAD TRAINING TIME, MEASURED ON EACH TARGET’S TRAINING SPLIT AND AVERAGED OVER THE 13 SETS (MEDIAN OF THREE REPEATS): IT IS THE COST THAT *does* VARY, WITH THE MATRYOSHKA DIMENSION (299 $\rightarrow$ 2176 MS FROM 64 TO 768) AND THE CLASSIFIER (485 MS FOR BALANCED LOGISTIC REGRESSION VS. 2217 MS FOR THE LINEAR SVM), AND IS SHOWN ON THOSE TWO AXES. IT IS 0 ON THE DETECTOR AND EMBEDDER AXES: THE VARIED COMPONENT ADDS NO PER-DEPLOYMENT TRAINING (DETECTORS ARE FROZEN/OFF-THE-SHELF AND THE EMBEDDER IS PRE-TRAINED), THE TYPING-HEAD FIT (≈ 2.2 S WITH THE DEFAULT BALANCED SVM@768) BEING CHARGED ON THE TYPING AXIS. THE EMBEDDER’S OWN OFFLINE FINE-TUNING (NONE FOR FROZEN NOMIC; ≈ 4 H FOR THE SELECTED HARD-MINING RUN) IS A SEPARATE ONE-TIME COST. $\P$ THE *nearest-centroid* (*zero-shot*) TYPING HEAD IS THE ONLY ONE THAT USES *no target labels*: IT ASSIGNS EACH MENTION TO THE NEAREST LABEL-NAME PROTOTYPE (TABLE VII); ITS FIT COLUMN IS THE ONE-OFF PROTOTYPE-EMBEDDING COST (293 MS), NOT A TRAINING PASS ON TARGET LABELS, WHICH IS WHY IT IS AN ORDER OF MAGNITUDE BELOW THE SUPERVISED HEADS WHILE REMOVING THE SUPERVISION OPENER OTHERWISE NEEDS.

Configuration	AMI	F ₁	Fit (ms)	p50 (ms)	En. (Wh)	CO ₂ (g)
<i>Mention detector</i>						
GLiNER-S	37.7	28.8	0.0	51	0.73	0.04
GLiNER-M	38.0	28.7	0.0	70	1.01	0.06
GLiNER-L	40.2	30.5	0.0	143	1.59	0.09
<i>Entity embedder*</i>						
frozen Nomic v1.5	25.8	43.9	0	42	0.63	0.04
contrastive (Triplet, CoNLL, 3 ep.)	54.0	63.3	0	42	0.63	0.04
+ hard-negative mining (2 ep., 3k tr.)	60.1	68.8	0	42	0.63	0.04
+ hard-negative mining (3 ep., 8k tr.)	62.5	71.8	0	42	0.63	0.04
<i>Matryoshka dimension$\S$</i>						
64	57.9	67.7	299	42	0.63	0.04
128	59.9	69.7	562	42	0.63	0.04
256	61.2	70.6	926	42	0.63	0.04
512	62.0	71.2	1556	42	0.63	0.04
768	62.5	71.8	2176	42	0.63	0.04
<i>Entity typing$\dagger$</i>						
GMM (per-class)	56.1	65.3	527	42	0.63	0.04
LogReg	55.6	53.5	546	42	0.63	0.04
LogReg (balanced)	57.8	67.5	485	42	0.63	0.04
LinearSVC	61.7	66.5	2041	42	0.63	0.04
LinearSVC (balanced)	62.5	71.8	2217	42	0.63	0.04
Nearest centroid (zero-shot) $\P$	44.4	46.8	293	42	0.63	0.04
+ transductive refine $\P$	51.4	48.5	293	42	0.63	0.04

TABLE VII

ENTITY TYPING ON GOLD MENTIONS: OPENER VS OWNER (AMI $\times 100$, $\uparrow$). DETECTION IS BYPASSED, ISOLATING THE TYPING COMPONENT. ABBREVIATIONS FOLLOW TABLE I; BOLD MARKS THE PER-DATASET WINNER. THREE SUPERVISION REGIMES ARE COMPARED. SUPERVISED OPENER FITS A LINEAR PROBE ON EACH TARGET’S TRAINING LABELS. OPENER-ZS[§] IS THE SAME PIPELINE MADE *zero-shot*: IT DROPS THE PROBE AND TYPES EACH MENTION BY COSINE SIMILARITY TO A LABEL-NAME PROTOTYPE (THE CENTROID OF THE LABEL’S ANCHOR WORDS EMBEDDED IN THE CONTRASTIVE SPACE), USING NO TARGET LABELS. OWNER[§] IS ZERO-SHOT AS WELL (CLUSTERING AFTER TRANSFER FROM CoNLL-2003). BOTH SHARE THE ERROR-DRIVEN HARD-NEGATIVE MINING EMBEDDER SELECTED IN TABLE VI. SUPERVISED OPENER LEADS ON ALL 13 SETS; AMONG THE TWO ZERO-SHOT SYSTEMS, OPENER-ZS EDGES OUT OWNER ON AVERAGE (44.4 VS 43.0) AT A FRACTION OF ITS INFERENCE COST (TABLE III, TABLE IV). OPENER-ZS (TRANSD.) ADDS A LABEL-FREE *transductive* REFINEMENT OF THE PROTOTYPES (SPHERICAL K-MEANS SEEDED BY THE LABEL NAMES, RUN ONCE ON THE UNLABELLED TEST MENTIONS), REACHING 51.4 AMI AND BEATING OWNER ON 11 OF 13 SETS; IT USES NO TARGET LABELS BUT, UNLIKE THE INDUCTIVE OPENER-ZS, EXPLOITS THE UNLABELLED TEST DISTRIBUTION. [†] CoNLL-2003 IS IN-DOMAIN FOR OWNER (ITS TRAINING SOURCE) AND FOR OPENER (ITS CONTRASTIVE SOURCE); IT IS NOT A TRANSFER RESULT AND IS NOT BOLD.

Model	AI	Lit	Mus	Pol	Sci	WNUT	Rest	Movie	Fab	Bio	CoNLL	GUM	GEN	Avg
OWNER [§]	55.7	54.3	59.8	63.2	63.1	30.5	29.3	31.8	31.9	29.2	52.5 [†]	30.0	27.2	43.0
OPENER-ZS [§]	41.4	61.4	55.2	50.7	58.0	33.0	42.6	18.4	35.6	54.1	65.9 [†]	31.1	30.2	44.4
OPENER-ZS (transd.) [§]	46.4	65.0	68.6	66.5	57.3	31.8	52.1	45.8	37.5	49.5	71.0 [†]	38.6	37.5	51.4
OPENER	57.6	67.0	71.8	74.8	68.9	38.6	63.0	66.9	59.2	71.2	71.5 [†]	50.3	51.5	62.5

TABLE VIII

ZERO-SHOT TYPING: EFFECT OF THE CONTRASTIVE EMBEDDER (OPENER-ZS, TYPING-ON-GOLD, MEANS OVER THE 13 SETS). THE NEAREST LABEL-NAME CENTROID HEAD IS HELD FIXED; ONLY THE EMBEDDER CHANGES. AMI, MACRO- F_1 AND ACCURACY ($\times 100$, $\uparrow$); BEST PER COLUMN IN BOLD. THE SELECTED HARD-NEGATIVE-MINING EMBEDDER IS FINE-TUNED ON THE TARGETS’ *training* SPANS (ENTITY-ENTITY), WHEREAS THE PILE-NER [10] EMBEDDERS ARE TRAINED ONLY ON PILE-NER ENTITY $\leftrightarrow$ LABEL-NAME PAIRS AND NEVER SEE ANY TARGET DATA, I.E. A STRICTER, FULLY DOMAIN-AGNOSTIC ZERO-SHOT. PILE-NER’S ≈ 4.6 K-TYPE VOCABULARY RAISES F_1 BUT NOT AMI, AND A SECOND LABEL-NAME HARD-MINING PASS DEGRADES BOTH BECAUSE PILE-NER’S GPT-GENERATED TYPES CONTAIN MANY NEAR-SYNONYMS (E.G. *person/per*) WHOSE MINED “CONFUSIONS” ARE HARMFUL; WE KEEP THE TARGET-TUNED EMBEDDER AND REPORT PILE-NER AS A NEGATIVE ABLATION. END-TO-END, THE PILE-NER VARIANT TIES THE SELECTED ONE (33.9 VS 34.2 AMI).

Embedder	Training data	AMI	F_1	Acc
hard-neg. mining (selected)	CoNLL + targets’ train	44.4	46.8	53.6
Pile-NER (entity $\leftrightarrow$ label)	Pile-NER only	42.2	50.6	53.9
+ label-name hard mining	Pile-NER only	39.7	41.4	44.7

V. CONCLUSION

[Placeholder - conclusion to be written.]

ACKNOWLEDGMENTS

[Placeholder - to be filled at submission time.]

REFERENCES

- [1] N. T. H. Nguyen, M. Miwa, and S. Ananiadou, "Span-based named entity recognition by generating and compressing information," in *Proceedings of the 17th Conference of the European Chapter of the Association for Computational Linguistics (EACL)*, 2023, pp. 1984–1996.
- [2] E. F. Tjong Kim Sang and F. De Meulder, "Introduction to the CoNLL-2003 shared task: Language-independent named entity recognition," in *Proceedings of the Seventh Conference on Natural Language Learning (CoNLL)*, 2003.
- [3] U. Zaratiana, N. Tomeh, P. Holat, and T. Charnois, "GLiNER: Generalist model for named entity recognition using bidirectional transformer," in *Proceedings of the 2024 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL-HLT), Volume 1: Long Papers*. Association for Computational Linguistics, 2024, pp. 5364–5376.
- [4] P. He, J. Gao, and W. Chen, "DeBERTaV3: Improving DeBERTa using ELECTRA-style pre-training with gradient-disentangled embedding sharing," in *The Eleventh International Conference on Learning Representations (ICLR)*, 2023.
- [5] Y. Ding, J. Li, P. Wang, Z. Tang, B. Yan, and M. Zhang, "Rethinking negative instances for generative named entity recognition," in *Findings of the Association for Computational Linguistics: ACL 2024*, 2024, pp. 3461–3475.
- [6] C. Raffel, N. Shazeer, A. Roberts, K. Lee, S. Narang, M. Matena, Y. Zhou, W. Li, and P. J. Liu, "Exploring the limits of transfer learning with a unified text-to-text transformer," *Journal of Machine Learning Research*, vol. 21, pp. 1–67, 2020.
- [7] P.-Y. Genest, P.-E. Portier, E. Egyed-Zsigmond, and M. Lovisetto, "OWNER: Toward unsupervised open-world named entity recognition," *IEEE Access*, vol. 13, pp. 50 077–50 105, 2025.
- [8] X. Wei, X. Cui, N. Cheng, X. Wang, X. Zhang, S. Huang, P. Xie, J. Xu, Y. Chen, M. Zhang, Y. Jiang, and W. Han, "ChatIE: Zero-shot information extraction via chatting with ChatGPT," *arXiv preprint arXiv:2302.10205*, 2023.
- [9] R. Cai, J. Lu, Z. Chen, B. Xu, and Z. Hao, "Handling missing entities in zero-shot named entity recognition: Integrated recall and retrieval augmentation," in *Proceedings of the 2025 Conference of the Nations of the Americas Chapter of the Association for Computational Linguistics: Human Language Technologies (NAACL)*, 2025, pp. 10 790–10 802.
- [10] W. Zhou, S. Zhang, Y. Gu, M. Chen, and H. Poon, "UniversalNER: Targeted distillation from large language models for open named entity recognition," in *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.
- [11] H. Touvron, T. Lavril, G. Izacard, X. Martinet, M.-A. Lachaux, T. Lacroix, B. Rozière, N. Goyal, E. Hambro, F. Azhar, A. Rodriguez, A. Joulin, E. Grave, and G. Lample, "LLaMA: Open and efficient foundation language models," *arXiv preprint arXiv:2302.13971*, 2023.
- [12] O. Sainz, I. García-Ferrero, R. Agerri, O. Lopez de Lacalle, G. Rigau, and E. Agirre, "GoLLIE: Annotation guidelines improve zero-shot information-extraction," in *The Twelfth International Conference on Learning Representations (ICLR)*, 2024.
- [13] Y. Tang, R. Hasan, and T. Runkler, "FsPONER: Few-shot prompt optimization for named entity recognition in domain-specific scenarios," in *Proceedings of the 27th European Conference on Artificial Intelligence (ECAI)*, 2024, arXiv:2407.08035.
- [14] S. Zhang, B. Cao, and J. Fan, "KCL: Few-shot named entity recognition with knowledge graph and contrastive learning," in *Proceedings of the 2024 Joint International Conference on Computational Linguistics, Language Resources and Evaluation (LREC-COLING)*. ELRA Language Resource Association, 2024, pp. 9681–9692.
- [15] S. S. Das, A. Katiyar, R. J. Passonneau, and R. Zhang, "CONTAiNER: Few-shot named entity recognition via contrastive learning," in *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2022.
- [16] Z. Nussbaum, J. X. Morris, B. Duderstadt, and A. Mulyar, "Nomic embed: Training a reproducible long context text embedder," *Transactions on Machine Learning Research (TMLR)*, 2025, published in TMLR 02/2025. OpenReview: <https://openreview.net/forum?id=IPmzyQSiQE>.
- [17] A. Kusupati, G. Bhatt, A. Rege, M. Wallingford, A. Sinha, V. Ramanujan, W. Howard-Snyder, K. Chen, S. Kakade, P. Jain, and A. Farhadi, "Matryoshka representation learning," in *Advances in Neural Information Processing Systems 35 (NeurIPS 2022)*, 2022.
- [18] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using siamese BERT-networks," in *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing (EMNLP)*, 2019.
- [19] F. Schroff, D. Kalenichenko, and J. Philbin, "FaceNet: A unified embedding for face recognition and clustering," in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2015.
- [20] G. Wang, C. Li, W. Wang, Y. Zhang, D. Shen, X. Zhang, R. Henao, and L. Carin, "Joint embedding of words and labels for text classification," in *Proceedings of the 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2018, pp. 2321–2331.
- [21] Y. Zhang, W. Chen, and L. Ma, "Contrastive learning with Gaussian embeddings and self-attention for few-shot named entity recognition," *Applied Sciences*, vol. 15, no. 21, p. 11819, 2025.
- [22] X. Hu, Y. Jiang, A. Liu, Z. Huang, P. Xie, F. Huang, L. Wen, and P. S. Yu, "Entity-to-text based data augmentation for various named entity recognition tasks," in *Findings of the Association for Computational Linguistics: ACL 2023*, 2023, pp. 9072–9087.
- [23] Y. Abbahaddou, F. D. Malliaros, J. F. Lutzeyer, A. M. Aboussalah, and M. Vazirgiannis, "Gaussian mixture models based augmentation enhances GNN generalization," 2024, arXiv:2411.08638 [cs.LG], preprint.
- [24] D. A. Reynolds, "Gaussian mixture models," in *Encyclopedia of Biometrics*, S. Z. Li and A. K. Jain, Eds. Springer US, 2009, pp. 659–663.
- [25] A. P. Dempster, N. M. Laird, and D. B. Rubin, "Maximum likelihood from incomplete data via the EM algorithm," *Journal of the Royal Statistical Society: Series B (Methodological)*, vol. 39, no. 1, pp. 1–22, 1977.
- [26] E. Strubell, A. Ganesh, and A. McCallum, "Energy and policy considerations for deep learning in NLP," in *Proceedings of the 57th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2019.
- [27] R. Schwartz, J. Dodge, N. A. Smith, and O. Etzioni, "Green AI," *Communications of the ACM*, vol. 63, no. 12, pp. 54–63, 2020.
- [28] D. Patterson, J. Gonzalez, U. Hölzle, Q. Le, C. Liang, L.-M. Munguia, D. Rothchild, D. So, M. Texier, and J. Dean, "The carbon footprint of machine learning training will plateau, then shrink," *Computer*, vol. 55, no. 7, pp. 18–28, 2022.
- [29] A. Lacoste, A. Luccioni, V. Schmidt, and T. Dandres, "Quantifying the carbon emissions of machine learning," *arXiv preprint arXiv:1910.09700*, 2019.
- [30] N. X. Vinh, J. Epps, and J. Bailey, "Information theoretic measures for clusterings comparison: Variants, properties, normalization and correction for chance," *Journal of Machine Learning Research*, vol. 11, pp. 2837–2854, 2010.
- [31] Z. Liu, Y. Xu, T. Yu, W. Dai, Z. Ji, S. Cahyawijaya, A. Madotto, and P. Fung, "CrossNER: Evaluating cross-domain named entity recognition," in *Proceedings of the AAAI Conference on Artificial Intelligence*, 2021.
- [32] L. Derczynski, E. Nichols, M. van Erp, and N. Limsopatham, "Results of the WNUT2017 shared task on novel and emerging entity recognition," in *Proceedings of the 3rd Workshop on Noisy User-generated Text (W-NUT)*, 2017, pp. 140–147.

- [33] J. Liu, P. Pasupat, S. Cyphers, and J. Glass, "Asgard: A portable architecture for multilingual dialogue systems," in *Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2013, pp. 8386–8390.
- [34] A. Kumar and B. Starly, "FabNER: Information extraction from manufacturing process science domain literature using named entity recognition," *Journal of Intelligent Manufacturing*, vol. 33, no. 8, pp. 2393–2407, 2022.
- [35] J.-D. Kim, T. Ohta, Y. Tsuruoka, Y. Tateisi, and N. Collier, "Introduction to the bio-entity recognition task at JNLPBA," in *Proceedings of the International Joint Workshop on Natural Language Processing in Biomedicine and its Applications (NLPBA/BioNLP)*, 2004, pp. 70–75.
- [36] A. Zeldes, "The GUM corpus: Creating multilayer resources in the classroom," *Language Resources and Evaluation*, vol. 51, no. 3, pp. 581–612, 2017.
- [37] T. Aoyama, S. Behzad, L. Gessler, L. Levine, J. Lin, Y. J. Liu, S. Peng, Y. Zhu, and A. Zeldes, "GENTLE: A genre-diverse multilayer challenge set for english NLP and linguistic evaluation," in *Proceedings of the 17th Linguistic Annotation Workshop (LAW-XVII)*, 2023.
- [38] J.-D. Kim, T. Ohta, Y. Tateisi, and J. Tsujii, "GENIA corpus—a semantically annotated corpus for bio-textmining," *Bioinformatics*, vol. 19, no. Suppl. 1, pp. i180–i182, 2003.
- [39] T. Dettmers, A. Pagnoni, A. Holtzman, and L. Zettlemoyer, "QLoRA: Efficient finetuning of quantized LLMs," *arXiv preprint arXiv:2305.14314*, 2023.
- [40] A. Paszke, S. Gross, F. Massa, A. Lerer, J. Bradbury, G. Chanan, T. Killeen, Z. Lin, N. Gimeshein, L. Antiga, A. Desmaison, A. Köpf, E. Yang, Z. DeVito, M. Raison, A. Tejani, S. Chilamkurthy, B. Steiner, L. Fang, J. Bai, and S. Chintala, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems 32 (NeurIPS)*, 2019.
- [41] T. Wolf, L. Debut, V. Sanh, J. Chaumond, C. Delangue, A. Moi, P. Cistac, T. Rault, R. Louf, M. Funtowicz, J. Davison, S. Shleifer, P. von Platen, C. Ma, Y. Jernite, J. Plu, C. Xu, T. Le Scao, S. Gugger, M. Drame, Q. Lhoest, and A. M. Rush, "Transformers: State-of-the-art natural language processing," in *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations (EMNLP)*, 2020.
- [42] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, J. Vanderplas, A. Passos, D. Cournapeau, M. Brucher, M. Perrot, and É. Duchesnay, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.
- [43] R.-E. Fan, K.-W. Chang, C.-J. Hsieh, X.-R. Wang, and C.-J. Lin, "LIBLINEAR: A library for large linear classification," *Journal of Machine Learning Research*, vol. 9, pp. 1871–1874, 2008.